# Singapore School Kinderland Secondary-2

## Computer Science (0478 / 2210)

### Term 2 • Academic Session 2026-27

### Class Test-2: Error Detection & Cryptographic Security (TB Pg 54–74)

**Date:** _____ **Student Name:** _____

**Total Marks: 25 Marks** **Marks Obtained:** _____

**Teacher's Signature:** _____ **Parent's Signature:** _____

## Section A: Fill in the Blanks using the Security Word Bank [ 5 Marks ]

**Word Bank: Parity Block • Check Digit • Checksum • Asymmetric • Automatic Repeat Request (ARQ)**

**1.** A _____ combines horizontal and vertical parity checks to locate and automatically correct a flipped bit.

**2.** A _____ is calculated on identification numbers (barcodes) to detect human transcription and transposition errors.

**3.** In _____ encryption, two mathematically paired keys (a Public Key and a Private Key) are utilized.

**4.** The _____ algorithm computes a numerical value from a block of data at sender and recalculates it at destination.

**5.** _____ is a transmission protocol that uses positive acknowledgements (ACK) and timeout timers to trigger retransmission.

## Section B: Write [ T ] for True or [ F ] for False [ 5 Marks ]

| No. | Technical Statement | Answer |
|---|---|---|
| 1. | If two bits flip in the same byte during transmission, a standard single parity check will fail to detect the error. | [ ] |
| 2. | In symmetric encryption, the sender and receiver use completely different, non-shared cryptographic keys. | [ ] |
| 3. | An Echo Check requires double the network bandwidth because the receiving computer sends all data back to the sender. | [ ] |
| 4. | Digital signatures are encrypted using the sender's private key to provide authenticity and prove non-repudiation. | [ ] |
| 5. | In an ARQ protocol, if a timeout occurs without receiving an ACK, the sender terminates transmission permanently. | [ ] |

## Section C: Match Cryptographic Concepts with Security Roles [ 4 Marks ]

| Security Concept | Role & Mechanism | Answer |
|---|---|---|
| 1. Public Key | A. Kept secret by owner; used to decrypt confidential messages | 1 → ( ) |

## Section D: Multiple Choice Questions (Cambridge Exam Pattern) [ 4 Marks ]

**1. What is the fundamental security flaw associated with Symmetric Encryption?**

(A) The shared secret key must be transmitted across the network, risking interception

(B) It can only encrypt numbers (C) Encryption speed is too slow

**2. If a customer mistypes an account number as "4589" instead of "4598", which error type is this?**

(A) Transposition Error (B) Transcription Error (C) Parity Error

**3. In an ARQ protocol, what packet does the receiver transmit if it detects a checksum mismatch?**

(A) Negative Acknowledgement (NAK) (B) Positive ACK (C) Timeout

**4. When user A sends an encrypted message to user B, which key is used to encrypt the plain text?**

(A) User B's Public Key (B) User A's Private Key (C) User A's Public Key

## Section E: Parity Block Matrix Analysis & ARQ Protocol Mechanics [ 7 Marks ]

**1. Parity Block Matrix Error Pinpointing Problem: (4 Marks)**

A transmission uses **Odd Parity**. The received block of bytes and its parity byte are shown below:

| Byte | b1 | b2 | b3 | b4 | b5 | b6 | b7 | Parity Bit |
|---|---|---|---|---|---|---|---|---|
| Byte 1 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 0 |
| Byte 2 | 0 | 1 | 1 | 0 | 1 | 0 | 1 | 1 |
| Byte 3 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 0 |
| Byte 4 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 0 |
| Parity Byte | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 |

a) Identify which byte row contains the corrupted parity count: _____

b) Identify which bit column has flipped: _____

c) State the corrected bit value: _____

**2. ARQ & Encryption Scenario Question: (3 Marks)**

Explain how the combination of **Timeout Timers** and **Negative Acknowledgements (NAK)** prevents infinite communication deadlocks when transmitting data over noisy satellite channels:

_____

_____

**Teacher's Assessment Feedback & Evaluation Rubric:**

| Assessment Criteria | Max Marks | Marks Awarded | Teacher Qualitative Remarks & Feedback |
|---|---|---|---|
| 1. Error Detection Methods & Check Digits | 8 | _____ | Strengths: _____ Focus Area: _____ Parent's Ack: _____ |
| 2. Cryptography, Hashing & SSL Certificates | 8 | _____ | |
| 3. Parity Block Solving & ARQ Protocol Mechanics | 9 | _____ | |
| **Total Marks Obtained** | **25** | _____ | **Grade: [ A* ] [ A ] [ B ] [ C ]** |

**Evaluator Signature:** _____ **Date:** _____